Eur Arch Otorhinolaryngol
DOI 10.1007/s00405-017-4452-8

REVIEW ARTICLE

Hearing loss in vitiligo: current concepts and review

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Received: 12 October 2016 / Accepted: 3 January 2017
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Abstract The objective of this study was to describe the occurrence, clinical manifestations, audiometric findings, pathogenesis and approach to sensorineural hearing loss (SNHL) among patients diagnosed with vitiligo with a review of the literature. We present a systematic review of the literature on cases of SNHL in patients diagnosed with vitiligo and studies conducted to investigate audiometric changes in such patients. Data on presentation, diagnosis and medical approach were reviewed. A total of 21 studies and case reports revealed at least 102 cases of SNHL in patients diagnosed with vitiligo. Arguments for a common causative etiology related to melanocyte function were mentioned in most of the literature. Evaluation of hearing function among all patients diagnosed with vitiligo seems to be an accepted approach; it should include audiometry, otoacoustic emissions (OAE) and ABR measurements. Extra precaution to prevent ototoxic or noise-induced hearing loss is strongly recommended. Further research is needed to better understand its pathogenesis.

Keywords Hearing loss · Vitiligo · Melanocytes

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Introduction

Vitiligo is a common progressive depigmentating skin disorder in which melanocytes of the skin are damaged. Destruction of melanocytes takes place within the epidermis, the mucous membranes, eyes and occasionally in some hair bulbs. Melanocytes of leptomeninges, retinal pigment epithelium, uveal tract and inner ear may also be involved. Vitiligo can appear at any age, but more often affects children and young adults, with the incidence of the disease declining with age. The disease prevalence is 0.5–1% in the general population with all ethnic groups affected. Vitiligo is characterized by the appearance of pearl-white macules of different sizes and shapes on skin and hair in the hypopigmentary area [1, 2]. It may be classified as localized, segmental, generalized and universal, according to the area of pigmentation loss. Functional melanocytes in patients with vitiligo disappear from the involved skin by a mechanism that is yet to be identified. Various theories have been proposed such as:

1. *auto-immunity* due to frequent associations with other auto-immune disease such as thyroiditis and type I diabetes, the presence of anti-melanocyte antibodies and a response to immunosuppressant therapy;
2. *cytotoxicity* the possibility that metabolites formed during melanin synthesis may destroy melanocytes;
3. *neural* chemical mediators released at nerve endings might destroy melanocytes or inhibit melanin production;
4. *free radicals* excess free radicals might be toxic for melanocytes;
5. *convergent* a combination of the above [2–4].

Published online: 14 February 2017

Recent clinical and experimental studies support the theory that pathogenetic mechanisms of vitiligo could be a systemic event, as vitiligo is associated with ocular and auditory abnormalities including sensorineural hearing loss (SNHL). Alphonse Corti (1831) was the first to mention the presence of pigment cells in the inner ear. Within the cochlea, melanin is found with lower relative amount in the basal area compared to the middle turn and apex [5]. However, the role of the melanocytes—and that of melanin—in the inner ear is not completely understood. Melanocytes are particularly present in the modiolus, osseous spiral lamina, Reissner's membrane and the vascular stria and in highly vascularized areas of apparently important secretory or metabolic function [6]. According to Savin, cells containing pigments partially or fully adhere to blood vessel walls, which are sites of intense metabolite exchanges. Therefore, it is probable that they have a vasomotor function in the inner ear. It is also known that they are necessary for the normal development and function of the stria vascularis [7–10]. Furthermore, melanocytes are involved in creating the endolymphatic potential and might act as free radical scavengers [7]. A loss of hearing in the higher frequencies may represent a decreased protective affect to free radicals by the reduced amount of melanocytes in the basal turn [5, 11, 12]. See Fig. 1.

Audiological abnormalities have been described in several hypopigmentary disorders such as albinism, Waardenburg syndrome, Vogt–Koyanagi–Harada syndrome and Alezzandrini syndrome. Carvalho found that otoacoustic emissions (OAE) and the high-frequency sensitivity auditory function were affected negatively in patients with pigment disorders [13]. Evidence of sensorineural hearing loss (SNHL) in vitiligo patients has also been reported over the last decade [14–20]. However, most patients with vitiligo are clinically asymptomatic for audiometric abnormalities. Other studies have not found such a relationship [21, 22].

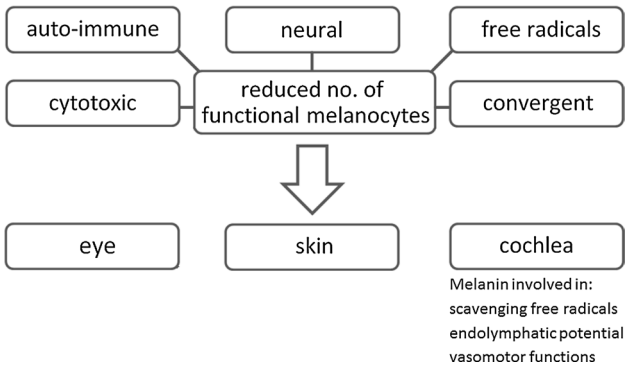

Fig. 1 Schematic overview of possible pathogenetic mechanisms of vitiligo in genetically predisposed patients. A reduced number of functional melanocytes in the cochlea may cause hearing loss by a reduced protective and supporting effect of melanin

Since audiology test results are conflicting, we therefore considered it worthwhile reviewing the current literature on SNHL in patients diagnosed with vitiligo and discuss the approach with such patients.

Materials and methods

Search strategy and selection

A systematic review of the literature was conducted using the databases of PubMed and the Cochrane Library on August 2016. We searched for articles relating to patients with vitiligo (domain) with audiology testing (intervention) and subsequently searched for articles relating to SNHL among those patients. Case reports as well as clinical studies if they contained a combination of domain and determinant were also searched. Articles were excluded if the main subject was not in relation to our domain in combination with determinant or in cases where the language was other than English. We screened references and related articles and conducted a general Internet search to verify if all significant articles were included until the day of admission of the manuscript. The results, acquired from the systematic literature review, will predominantly be presented in the tables. Case reports are presented in Table 1, but not further mentioned in the results since the data were often incomplete. Most of those articles appear in the literature between 1972 and 1992, the same period in which autoimmune SNHL was first mentioned [23].

We assessed and compared the hearing performances based on different audiology tests in patients with vitiligo in the studies. In addition to the hearing performance, we evaluated the different hypotheses about the mechanism responsible for hearing loss among vitiligo patients and proposed a treatment strategy for SNHL possibly related to vitiligo.

Results

Pure tone audiometry thresholds

More accurate descriptions of audiology testing among vitiligo patients appear in controlled clinical studies from 1991 until the present day. Eight studies were found evaluating hearing thresholds in vitiligo patients and healthy controls based on pure tone audiometry. Although there was no SNHL found in the study first described by Escelante et al., all other studies reported a higher prevalence of SNHL among patients with vitiligo, ranging from 4.0 to 68.8%, compared to healthy controls [16, 18, 20, 22, 29–31]. Also in the study by Hong et al., significant lower

Table 1 Case reports and retrospective studies describing hearing loss among patients with vitiligo (1997–1994)

Study (year of publication)	Main outcome
Thurmon et al. (1976) [24]	Seven cases of congenital deafness and an autosomal recessive pedigree pattern were observed in inbred kindred. Three of the cases also had vitiligo, and there were two other cases of vitiligo in this kindred
Kumakiri et al. (1982) [25]	A 35-year-old Japanese woman developed vitiligo with an inflammatory erythema on the face 2 years after the onset of bilateral uveitis and dysacusis
Tosti A et al. (1987) [14]	The audiologic function in 50 patients affected by vitiligo was studied. Eight patients (16%) had sensorineural hearing loss at presentation
Dereymaeker et al. (1989) [26]	Two non-related patients (<30 years) presented with retinitis pigmentosa, hearing loss and vitiligo
Nishida et al. (1990) [27]	A 48-year-old female patient with vitiligo developed primary hypothyroidism and hypoacusis. The signs and symptoms of hypothyroidism improved with thyroid hormone replacement therapy, but hypoacusis did not
Cernea and Damien (1994) [28]	Observation of a 47-year-old patient with pigmentary retinopathy, hypoacusis and vitiligo A certain parallelism between the degeneration of the retineal epithelium and the Corti neurosensorial epithelium was suspected

thresholds were found in Asian vitiligo patients compared to their controls [19].

Most of them had a threshold of >30 dB HL. The high prevalence of SNHL in the study of Fleissig et al. included mild and minimal SNHL as well. In more recent studies, attention was given to the thresholds at different frequencies. Significantly higher (worse) thresholds were found among vitiligo patients, at frequencies between 4 and 8 kHz with P values <0.05. To a lesser extent, there were higher thresholds among vitiligo patients for 0.125, 1, 10 and 12.5 kHz. The test results are summarized in Table 2 [16, 18–22, 29–33]. Fleissig et al. compared the shapes of the audiograms from 16 vitiligo patients and 16 healthy controls. They found a significantly more common "notch shape" audiogram in the higher frequencies for vitiligo patients, with no history of noise-induced hearing loss ($P = 0.021$). Very high frequencies (over

Table 2 Prevalence (in percentage) of SNHL among patients and controls during audiology testing; and P values for significant higher pure tone thresholds among vitiligo patients and the related frequencies

P value (for different prevalence in patients and controls)	SNHL found during audiometry (%)	Frequencies tested (kHz)	Number of subjects (n)	Study (year of publication)
0.01	0% 0%	0.125–8 kHz	47 patients 35 controls	Escelante-Ugalde et al. (1991) [21]
No P value	4% 0%	0.25–8 kHz	50 patients 50 controls	Ozuer et al. (1998) [22]
<0.05	No percentage of SNHL within groups	4–10 kHz	29 patients 41 controls	Ardic et al. (1998) [16]
0.006	14% 0%	2–8 kHz	57 patients 50 controls	Aydogan et al. (2006) [18]
<0.005	20% 2%	0.250–8 kHz	150 patients 100 controls	Gopal et al. (2007) [29]
No P value	12.5%	0.250–8 kHz	24 patients	Angrisani et al. (2009) [32]
$P < 0.05$	– –	1–8 kHz	89 patients 89 controls	Hong et al. (2009) [19]
$P < 0.05$	36.4% 6.8%	0.25–12.5 kHz	22 patients 22 controls	Aslan et al. (2010) [20]
$P > 0.05$	21% 15%	0.25–8 kHz	197 patients 100 controls	Al-Mutairi et al. (2011) [31]
P 0.023	68.8% 6.3%	0.25–12 kHz	16 patients 16 controls	Fleissig et al. (2013) [30]
No P value	38.1% –	0.25–8 kHz	21 patients 20 controls	Mahdi et al. (2012; 2016) [33]

SNHL sensorineural hearing loss

8 kHz) were tested in three studies and found to be significantly higher in the vitiligo population [16, 20, 30].

Auditory brainstem evoked responses

An impression of cochlear and retrocochlear performance can be gained from auditory brainstem evoked responses (ABR). Five studies used ABR to click stimuli between 100 and 150 and 3000 Hz. While Ozuer et al. did not find any significant difference between 50 vitiligo patients and 50 healthy controls, both Nikifroidis et al. and Hong et al. reported a decrease in wave I peak latency and an increased I–III interpeak latency ($P<0.01$ and $P<0.05$, respectively) [15, 19, 22]. Additionally, patients with active disease had a significant decrease in peak wave I latency and significant increases in interpeak I–III and interpeak I–V latencies compared to patients with a stable disease in the study by Hong et al. [19]. Two studies reported no significant difference in wave I peak latency, but both wave III peak latency and I–III interpeak latency were significantly increased in both ears of vitiligo patients compared to healthy controls ($P=0.02$ and $P=0.008/P=0.001$, respectively) [18, 33].

Otoacoustic emissions

Three prospective studies included transient OAE in their evaluation to find possible cochlear dysfunction [20, 30, 32]. They all reported on frequencies of 1–4 kHz with 70 or 75% reproducibility and a response criteria of three frequencies at 3 dB or greater above the noise floor. Interestingly, they all found failure only at higher frequencies, 3 or 4 kHz. Two studies found significantly reduced otoacoustic emission amplitude at 4 kHz [20, 32]. All healthy controls in the studies passed the test.

In addition, Angrisani et al. reported that in a third of the cases they examined, suppression of OAEs was absent although white noise was presented to the contralateral ear (to test the function of the efferent pathways). 66.7% of the vitiligo patients with normal audiometry results had partial or total loss, among them 64.7% with bilateral failure mostly at the high frequencies (63.3%) [32]. Fleissig et al. found only 2 out of 16 patients who failed [30]. Aslan et al. found significantly reduced otoacoustic emission amplitude at 4 kHz [20]. All healthy controls in the studies passed the test. Interestingly Anbar et al. examined 53 patients with vitiligo and normal audiograms, 60% of patients had cochlear dysfunction based on DPOAEs, often bilateral, while their control group had no abnormality [34].

In addition, Angrisani et al. reported that in a third of the cases they examined, suppression of OAEs was absent although white noise was presented to the contralateral ear (to test the function of the efferent pathways) [32].

Discussion

Several case reports and studies have been published mentioning SNHL in patients with vitiligo. Some suggest a systemic mechanism responsible for both the symptoms of depigmentation in vitiligo and the hearing loss. Although several theories have been proposed, the exact pathogenesis remains unclear. A study by Sun et al. reported that Afro-American individuals had greater cochlear melanin levels compared to Caucasian individuals and overall better hearing thresholds [35]. To our knowledge, no research has been conducted to investigate the presence or absence of melanocytes in the ears of patients with vitiligo compared to healthy controls. The first studies evaluating hearing performance among vitiligo patients based on audiology tests did not confirm a significantly higher prevalence of SNHL in this specific group compared to healthy controls. However, they compared only pure tone audiometry at 0.5–2KHz or ABR, whereas later studies found that the hearing loss among patients diagnosed with vitiligo presented in the higher frequencies.

More recently, a larger number of clinically controlled studies did confirm the hypothesis that SNHL is significantly more common among vitiligo patients, between 4.0 and 68.8% of patients. The higher percentage includes also very mild SNHL and was described by Fleissig et al. [30]. They also found a negative relation to mucosal onset of disease and for patients more than 20 years old. In other studies, variables such as the duration of disease, gender, the age of the patient and the type of vitiligo did not correlate with the hearing loss. It should be noted that only one study mentioned a clear definition of sensorineural hearing loss as a difference ≥25 dB HL in at least two consecutive frequencies [20].

Audiometric testing of patients diagnosed with vitiligo demonstrated in most cases a hearing loss in the higher frequency range, 3–4 KHz during pure tone audiometry. Fleissig et al. described a typical notch-shaped audiogram that was more frequent among vitiligo patients and showed similarity to the audiogram common in noise-induced hearing loss [30]. The concept of noise-induced hearing loss in pigmentation disorders has previously been published in an experimental animal study in 1986 by Conlee et al. [11]. They found a reduced hearing restoration in albino guinea pigs after noise exposure compared to a control group. It seems that acoustic trauma appears to be of greater functional significance in strial dysfunction and thus makes vitiligo patients even more prone to hearing loss after noise exposure.

ABR measurements in patients with vitiligo by Nikifroidis et al. showed a decrease in W I peak latency and an increased I–III interpeak latency, which was later confirmed in a study carried out by Hong et al. [15, 19]. The

latency of the first wave is affected by interconnections of cochlear mechanisms, the movement of the basilar membrane and displacement of hair cells. The interpeak latency of wave I–III represents the action potential, synaptic activity of the otoacoustic nerve and conduction at the superior olivary nucleus. Whereas melanocytes are known to be involved in generating the endolymphatic potential, the role of melanocytes in retrocochlear conduction remains unclear.

A third hearing test performed among patients diagnosed with vitiligo was transient OAE [20, 30, 32]. Whereas 100% of healthy controls in all studies showed responses, a variable percentage of vitiligo patients showed partial or complete failure mostly in the higher frequency range. Interestingly, some patients showed abnormal transient OAE, but had a normal audiogram. Therefore, the prevalence of cochlear impairment may be higher among vitiligo patients than found only by audiogram evaluation. In addition, the lack of suppression in some patients indicates the possibility of damage to the medial olivocochlear bundle [32].

Conclusions

Since SNHL seems to be more common among vitiligo patients with hearing loss in the high-frequency range, audiometry should be performed including very high frequencies such as 10 and 12 kHz. As there are indications of impaired cochlear function with normal audiometry results, testing of hearing function with OAE is recommended. Suppression of OAE should be attempted to discover abnormal auditory efferent function. ABR should be performed, as abnormalities have been reported in vitiligo patients. Furthermore, since there are no variables such as age, gender or type of vitiligo that correlate to hearing loss, all patients, including those who are clinically asymptomatic, should be tested and preferably at first presentation. Further studies are necessary to determine the recommendations for interval and frequency of follow-up. Vitiligo patients that present with sensorineural hearing loss of sudden onset should be treated by steroidal therapy; either by oral intake or intratympanic injection of steroids.

In summary, audiological evaluation of patients with vitiligo should include audiometry and very high-frequency audiometry, OAE and ABR measurements. Incomplete evaluation might result in underdetection of hearing loss. Additionally, patients should take extra precautions to prevent acoustic trauma and avoid ototoxic drugs, especially during active disease, as their auditory system may be less protected due to nonfunctional melanocytes.

To fully understand the mechanism by which melanocytes function in relation to hearing, we recommend further research among vitiligo patients to gain a better understanding of the audiology functions and cell mechanisms in the absence of functional melanocytes in the ear.

Compliance with ethical standards

Conflict of interest The authors report no financial interests or potential conflicts of interest.

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